Maxwell Peng

maxwellpeng88@gmail.com | (647) 562-2398 | linkedin.com/in/maxwell | github.com/maxwell-penguin | maxwellpeng.com

Education

Western University — London, ON
B.Sc. Computer Science (Honours Specialization) — *Expected 2029*

- **GPA**: 4.0/4.0 | Western Scholarship of Distinction | EY x Accenture Case Competition Finalist
- **Relevant Coursework**: Data Structures & Algorithms, Software Tools & Systems Programming, Software Engineering, Computer Organization & Architecture, Discrete Structures

Technical Skills

Languages: Python, TypeScript, JavaScript, Go, Java, C++, SQL.
Backend: FastAPI, Node.js, RESTful APIs, WebSockets, OAuth, async pipelines, event-driven architecture.
AI/ML: LangGraph, OpenAI Agents SDK, multi-agent orchestration, LLM pipelines, Cerebras, scikit-learn.
Frontend: React, Next.js, Tailwind, Zustand, TypeScript, data visualization, responsive UI.
Databases: PostgreSQL (schema design, indexing, query optimization), Redis (KV/caching), data modeling.
Tools & Infra: Docker, GitHub Actions, CI/CD, AWS, Vercel, Railway, pytest, Playwright, Git.

Experience

Resilient Privacy — August 2026 – Present
Backend Engineering Intern — *Dallas, TX*

- Built a Go pipeline detecting company tech stacks from HTML, headers, and scripts via **13** validated fingerprint rules.
- Designed a **5**-category stakeholder classification engine with confidence scoring and testable HubSpot/Salesforce connectors.
- Shipped an outreach recommendation engine to production with success, partial, and error logging on every sync run.

Autumn — March 2026 – August 2026
Software Engineer Intern — *London, ON*

- Cut onboarding time from 20 to **12 min** with a Puppeteer scrape-to-prefill system, magic-link resume, and async emails.
- Reduced provider drop-off by **40%** with a 27-screen Next.js/TypeScript flow and a real-time completion scoring engine.
- Owned the REST API and PostgreSQL models end to end, with integration tests and automated CI/CD.

Dice AI — May 2025 – June 2026
Cofounder & Software Engineer — *Oakville, ON*

- Built a fintech platform for creators from 0 to 1 with a double-entry ledger so every amount traces to its source.
- Shipped **10+** OAuth integrations (YouTube, Instagram, Stripe, TikTok), cutting manual data entry by **90%**.
- Built an AI contract review pipeline with PII sanitization, structured output validation, and retry logic.

Projects

SketchStorm Studio | *FastAPI, WebSockets, React/TypeScript, Tailwind, Docker, Playwright* — 2026

- Built a **6-agent** AI pipeline at the **Cerebras x Google DeepMind Hackathon** that turns a hand-drawn sketch into a working React/Tailwind app in real time. **15,000+** lines, live at cerebrashacks.vercel.app.
- Six single-purpose agents (Vision Parser, Architect, Code Forge, Auditor, Accessibility Checker, Vision Critic) coordinated over WebSockets, with the critic comparing output back to the original sketch.
- Shipped **26** passing tests (fault isolation, agent retries, WebSocket schema validation) plus Playwright visual QA that uses headless Chromium screenshot diffing to catch regressions.

Vigil | *Go, React/Vite/Tailwind, SQLite, GitHub API, Python/JS SDKs* — 2026

- Built an open-source SRE toolkit in Go that alerts on SLO burn rate using the Google SRE Workbook's two-window model, with configurable thresholds and alert suppression to cut false positives.
- Automatically opens a postmortem GitHub Issue when an SLO is breached, so every incident gets a written record.
- Added a live dashboard (React/Vite/Tailwind, SQLite) and Python/JS SDKs so other services can report their own metrics.

Defrost | *Next.js, TypeScript, Cohere Embed v4, Claude API, OCR* — 2026

- Built a tool that reads a company's About page or screenshot, matches it to my most relevant project with Cohere Embed v4, and drafts a cold email in my voice, replacing a manual step in every application.
- Chose embeddings over keywords since companies describe the same problem differently; added OCR for screenshots.
- Used few-shot prompting on my real outreach emails so drafts sound like me, not generic AI. I use it daily.

Interests

Athletics: **Team Canada Badminton** (PanAm Championship); Chess (Elo: **1936**).
Leadership: Residence Don — Western University; Software Engineer — Western Founders Network (700+ users).
Research: Undergraduate Research Assistant, CulturePlex Lab, Western University (Fall 2026).